

Zed Block Examples

Example 1 : Basic Variable Declarations

Use axdef for variable declarations with types:

$[Person, Department]$

$assignment : Person \rightarrow Department$ $employees : \mathbb{F} Person$ $departments : \mathbb{F} Department$
--

This groups related variable declarations in a single block (given types must be declared outside).

Example 2 : Abbreviations

$N_PLUS == \{ n : \mathbb{N} \mid n > 0 \}$
 $N_EVEN == \{ n : \mathbb{N} \mid n \bmod 2 = 0 \}$
 $N_ODD == \{ n : \mathbb{N} \mid n \bmod 2 = 1 \}$

Abbreviations can be defined standalone (zed blocks for abbreviations may not be fully supported yet).

Example 3 : Type Parameters

$[Company]$

Declare given types using the given keyword.

Example 4 : Predicates elem Zed Block

$x : \mathbb{N}$ $y : \mathbb{N}$
$x > 0$ $y > 0$ $x + y < 100$

Multiple predicates can constrain variables defined in axdef.

Example 5 : Free Type Definitions

$Color ::= red \mid green \mid blue$
 $Size ::= small \mid medium \mid large$

$Product$
$color : Color$ $size : Size$ $price : \mathbb{N}$
$price > 0$

Free types are defined standalone, then used in schemas.

Example 6 : Multiple Schemas

[*AccountZB*, *BalanceZB*]

<i>BankAccountZB</i>
<i>accountNumber</i> : $\mathbb{N}$ <i>balance</i> : $\mathbb{Z}$
<i>balance</i> ≥ 0

<i>TransactionZB</i>
<i>from</i> : <i>BankAccountZB</i> <i>to</i> : <i>BankAccountZB</i> <i>amount</i> : $\mathbb{N}$
<i>amount</i> > 0 <i>from.balance</i> $\geq$ <i>amount</i>

Multiple schemas can be defined separately, each with their own constraints.

Example 7 : Schema with External Variables

[*Student*, *Course*]

<i>enrolled</i> : <i>Student</i> $\leftrightarrow$ <i>Course</i> <i>maxCourses</i> : $\mathbb{N}$ <i>maxStudents</i> : $\mathbb{N}$
<i>maxCourses</i> = 7 <i>maxStudents</i> = 500

<i>Enrollment</i>
<i>students</i> : $\mathbb{F}$ <i>Student</i> <i>courses</i> : $\mathbb{F}$ <i>Course</i>
<i>students</i> = dom <i>enrolled</i> <i>courses</i> = ran <i>enrolled</i> $\#students \leq maxStudents$ $\#courses \leq maxCourses$

Schemas can reference variables defined in axdef blocks.

Example 8 : Nested Schemas

[*Char*]

<i>Address</i>
<i>street</i> : seq <i>Char</i> <i>city</i> : seq <i>Char</i> <i>zipCode</i> : $\mathbb{N}$
<i>zipCode</i> $\geq 10000 \wedge zipCode \leq 99999$

$ \begin{array}{l} \textit{PersonData} \\ \textit{name} : \text{seq } \textit{Char} \\ \textit{address} : \textit{Address} \\ \textit{age} : \mathbb{N} \end{array} $
$\textit{age} \geq 0 \wedge \textit{age} \leq 150$
$ \quad \textit{population} : \mathbb{F} \textit{PersonData}$

Schema *Address* is used within schema *PersonData*. Variables using schemas go in axdef.

Example 9 : Constants and Functions

$PI == 3$

$E == 2$

$ \begin{array}{l} \textit{circumference} : \mathbb{N} \rightarrow \mathbb{N} \\ \textit{area} : \mathbb{N} \rightarrow \mathbb{N} \end{array} $
$ \begin{array}{l} \forall r : \mathbb{N} \bullet \textit{circumference}(r) = 2 * PI * r \\ \forall r : \mathbb{N} \bullet \textit{area}(r) = PI * r * r \end{array} $

Mathematical constants defined as abbreviations, functions in axdef.

Example 10 : Permission System Example

$[UserID, DocumentID]$

$\textit{PermissionType} ::= \textit{read} \mid \textit{write} \mid \textit{admin}$

$ \begin{array}{l} \textit{User} \\ \textit{userId} : \textit{UserID} \\ \textit{documents} : \mathbb{P} \textit{DocumentID} \\ \textit{permissions} : \textit{DocumentID} \rightarrow \textit{PermissionType} \end{array} $
$\text{dom } \textit{permissions} \subseteq \textit{documents}$
$ \begin{array}{l} \textit{System} \\ \textit{users} : \textit{UserID} \rightarrow \textit{User} \\ \textit{allDocuments} : \mathbb{F} \textit{DocumentID} \end{array} $
$\textit{allDocuments} = \bigcup \{ u : \text{ran } \textit{users} \bullet u.\textit{documents} \}$

$| \quad \textit{users} : \textit{UserID} \rightarrow \textit{User}$

A permission system specification using free types and schemas.

Example 11 : Best Practices

Use zed blocks for: 1. Type parameter declarations $[X, Y]$ 2. Abbreviations (N_PLUS , N_EVEN , etc.) 3. Simple predicates without variable declarations

Use axdef for: 1. Variable declarations with types 2. Constraints on those variables 3. Function definitions

Use schema for: 1. State spaces with declarations 2. Operations on state 3. Reusable component specifications